Seat No.:_____

AHU ID No._____ Name_____ Major_____ Grade_____

装 订 线

Stony Brook Institute at Anhui University

2022 Fall Semester

Applied Calculus // Final Exam (Paper A)

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	Total score
Score					
Grader					

I. Blank Fillings (5 items, 3 points each)

Score	

1. $\lim\limits_{n\to\infty}\left(\frac{1}{n^2}+\frac{2}{n^2}+\cdots+\frac{n-1}{n^2}\right)=$_____.

2. Let $f(x)=\int_{\frac{1}{2}}^{x}\ln t dt$ $x\in\left[\frac{1}{2},\frac{3}{2}\right]$, then the minimum value of the function can be obtained at $x=$_____.

3. The area bounded by the curve $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>0,b>0)$ is _____.

4. The domain of $\Gamma(x)=\int_0^{+\infty}t^{x-1}e^{-t}dt$ is _____.

5. The general solution of the equation $\frac{dy}{dx}+p(x)y=0$ is _____.

II. Choices (5 items, 3 points each)

Score	

6. If $f(x)$ is continuous on $[0,1]$ and $F'(x)=f(x)$, then $\int_0^1 f(2x)dx$ is ( )

A. $F(1)-F(0)$. B. $F(2)-F(0)$.

C. $\frac{1}{2}(F(2)-F(0))$. D. $2(F(2)-F(0))$.

7. Let $f(x)$ be continuous on $[0,a]$ and $f(x)+f(a-x)\neq 0, x\in[0,a]$, then the integral

$\int_0^a\frac{f(x)}{f(x)+f(a-x)}dx$ is ( )

A. 0.　　B. $\frac{a}{2}$.　　C. 1.　　D. a.

8. Suppose $f(x)$ possesses second derivative on $[0,1]$, and $f(x) \geq 0$, $f''(x) > 0$, then ( )

A. $\int_0^1 f(x)\,dx < \frac{f(0)+f(1)}{2}$.　　B. $\int_0^1 f(x)\,dx > \frac{f(0)+f(1)}{2}$.

C. $\int_0^1 f(x)\,dx = 0$.　　D. $\frac{f(0)+f(1)}{2} < f\left(\frac{1}{2}\right)$.

9. The convergent improper integral is ( )

A. $\int_0^{+\infty} \frac{1}{1+x}dx$.　　B. $\int_0^{+\infty} \frac{x}{1+x^2}dx$.　　C. $\int_0^{+\infty} \frac{1}{\sqrt{x}}dx$.　　D. $\int_0^{+\infty} \frac{1}{1+x^2}dx$.

10. In the following, the unique 1-order linear differential equation is ( )

A. $\frac{dy}{dx} + \tan x \cdot \sin y = \cos x$.　　B. $\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2 = x$.

C. $\frac{dy}{dx} - xy^2 = 0$.　　D. $\cos x \cdot \frac{dy}{dx} = \sin x$.

III. Calculations (8items, 8 points each)

Score	

11. Assume $f(x)$ has continuous derivative and $f(x) > 0, \ln f(x) = \sin x$, find $\int \frac{xf'(x)}{f(x)}dx$.

12. Find $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\sin x}{1+x^2} + \sin^2 x\right)dx$.

13. Calculate the area bounded by the curves $y^2 = 2x$ and $y = x - 4$.

14. Let l be a tangent line of the curve $y = x^2 \left(x \geq 0\right)$ with the tangent point M, A be the part bounded by the curve, l and the x-axis, the area of A is $\frac{1}{12}$, find

(1) The point M and the tangent line l;

(2) The volume of the revolution of A about x-axis;

15. Find the length of the curve $\begin{cases} x = a\left(\theta - \sin\theta\right) \\ y = a\left(1 - \cos\theta\right) \end{cases} \left(0 \leq \theta \leq 2\pi\right)\left(a > 0\right)$.

16. Let V be the volume of the revolution of $y=\frac{\sqrt{x}}{1+x^2}$ about x-axis, $V(\xi)$ be the part between $x=0$ and $x=\xi(\xi>0)$, calculate $\lim\limits_{\xi\to+\infty} V(\xi)$.

17. Find the general solution of the differential equation $(x-y-1)dx+(4y+x-1)dy=0$.

18. Find the general solution of the differential equation $y'+\frac{1}{x}y=\frac{\sin x}{x}$.

IV. Modelling Problem (6 points)

Score	

19. Assume that the growth rate of the bacteria number is positive proportional to the number of bacteria $x(t)$, and the ratio coefficient is $k > 0$. The number of bacteria was counted every cycle. The record shows that the number of bacteria reaches 150 units after three cycles, where the initial number of bacteria is 10, find the expression of the number of bacteria $x(t)$.